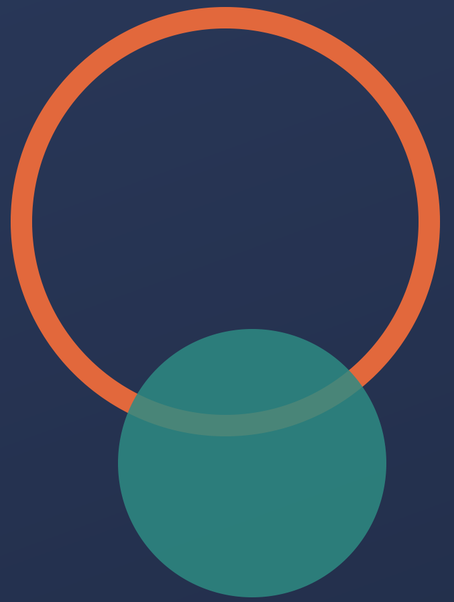
IDA117V GROUP PROJECT

Seeing municipal audit failure coming

Why we chose this problem, what our product solves, and how it works

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Lecturer: Dr Dan Masethe · 20 October 2026



THE PROBLEM

Municipal finances fail quietly, and the audit tells us too late

Poor financial management: weak cash, low debt collection, irregular spending

→

Unpaid suppliers and neglected infrastructure

→

Water, electricity and refuse services fail for residents

→

The audit result arrives after the year closes, so support comes after the damage

45.5% of the 1,434 municipality-years in our data were followed by a poor audit outcome the next year: qualified, adverse, disclaimer or outstanding.

WHY WE CHOSE THIS PROJECT

Four reasons this problem was worth solving

1 **A real service-delivery problem**

Audit outcomes show whether a municipality can manage the money behind basic services.

2 **A clear decision-maker**

National Treasury and the provincial treasuries decide each year where scarce support goes.

3 **Authoritative, open data**

Treasury’s Municipal Money covers every municipality, is free to reuse and holds no personal data.

4 **A measurable outcome**

The Auditor-General publishes an opinion every year, so every prediction can be checked against reality.

WHAT IT SOLVES

Four problems Treasury faces today

PROBLEM	WHAT OUR PRODUCT DOES
Support arrives after the damage	→ Forecasts each municipality’s next audit from figures already published
Information is scattered	→ One profile per municipality: audit history, finances, peers and norms
Prioritisation is inconsistent	→ The same documented risk score and ranking for all 245 municipalities
Decisions are hard to explain	→ Every score comes with plain-language reasons and Treasury norms

OUR SOLUTION

A forecast of every municipality’s next audit, and a dashboard to act on it

245

municipalities forecast for the 2025 audit

106

flagged at risk, 102 of them at 80% or more

90.8%

correct on a year the model never saw

HOW IT WORKS

From public data to a decision in five steps

1 · Collect

Audited figures and Auditor-General opinions from Treasury's Municipal Money API, 2018–2024

2 · Engineer

Nine financial-health ratios plus last year's audit outcome for each municipality and year

3 · Learn

Six models compared on past years; the simplest reliable one, Logistic Regression, is locked in

4 · Forecast

The locked model scores every municipality's next audit, with the reasons behind each score

5 · Act

A dashboard where officials drill from the national picture to one municipality

The whole pipeline reruns with one command, **python run_all.py**, and gives identical results every time.

WHY WE BUILT IT THIS WAY

Five design choices, and the reason for each

Choice	Why
Predict next year’s audit	Treasury needs time to act before the outcome is decided
Split the data by year	Learn from the past, test on the future: exactly how it will be used
Logistic Regression	As accurate as the neural networks, more stable, and every score can be explained
Macro-F1 as the main score	A missed failing municipality and a false alarm both have real costs
A dashboard, not a spreadsheet	Officials move from a national view to one municipality in two clicks

DOES IT WORK?

Tested once, on a year it had never seen

90.8%
of 240 municipalities classified correctly

90/103
municipalities that failed were flagged in advance

9/137
healthy municipalities wrongly flagged

0.906
macro-F1, 95% interval 0.87 to 0.94

A fair test

Trained on 2018–2022, then scored once on 2023 figures against the published 2024 audits.

Checked statistically

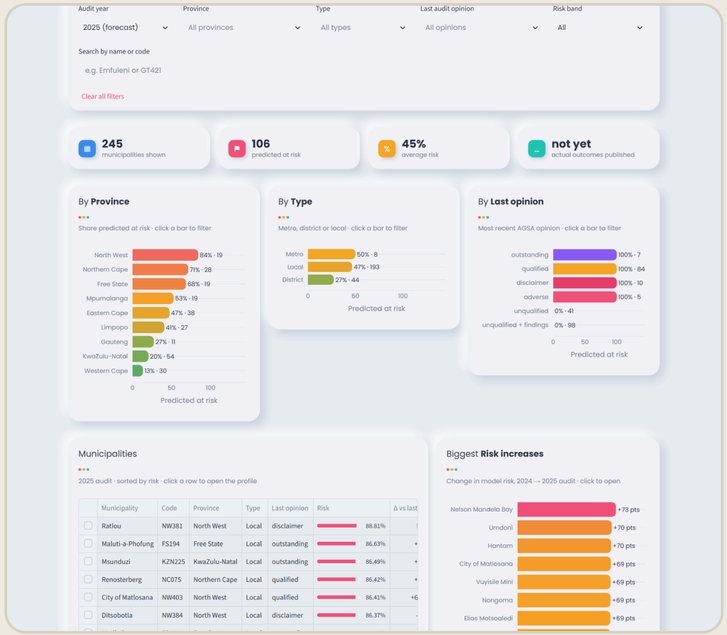
Compared with four other models, ablation tests and McNemar’s test; results reproduce exactly.

THE PRODUCT

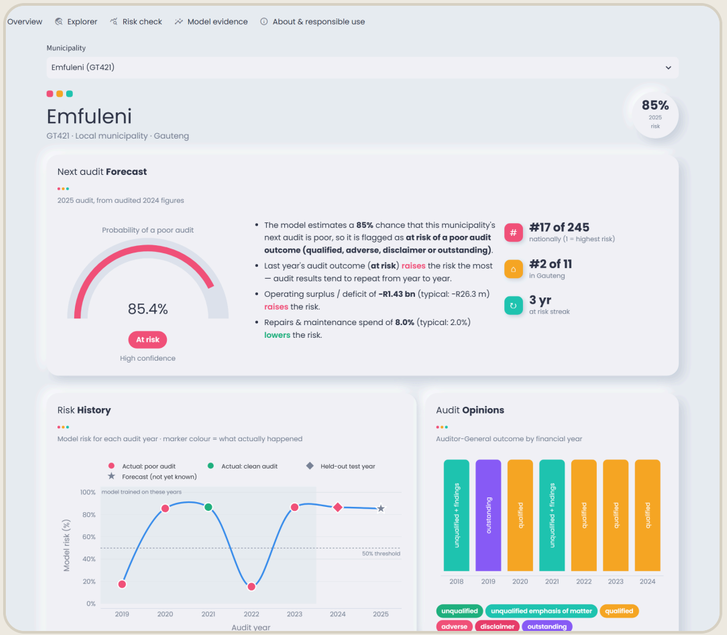
A dashboard built for the people who decide



Overview: the national picture; every number opens its detail



Explorer: filter by province, type, opinion or risk



Profile: one municipality's forecast, reasons and history

IN PRACTICE

How a Treasury official would use it

1	**See who is at risk**	The overview lists the 106 municipalities flagged for the 2025 audit
2	**Narrow to a province**	The explorer filters the list for each provincial treasury
3	**Understand one municipality**	The profile shows why it is flagged, its history and where it misses Treasury's norms
4	**Decide the support**	The official, not the model, chooses secondments, oversight or other help
5	**Check and retrain yearly**	When the Auditor-General publishes, compare the forecast and retrain

BEING HONEST ABOUT THE LIMITS

It knows which municipalities are struggling, not which are about to change

0 of 22

municipalities whose audit status changed were predicted correctly. Audit outcomes mostly repeat, and the model learned exactly that.

28.6%

false alarms in the Northern Cape, against 5–10% in other provinces: lists need a province-by-province review.

1,271%

irregular expenditure reported by one municipality: self-reported figures contain errors, and the worst performers submit the least data.

So we recommend it as a **triage list** for municipalities already in difficulty, never as an automatic funding or penalty tool.

WHAT COMES NEXT

Turning a triage list into an early warning

In-year data

Quarterly section 71 reports show trouble months before year-end.

Richer signals

Audit findings, senior vacancies and debt-collection trends.

Model the change

Predict whether a status will change, not just what it will be.

Thank you. Questions?

Dashboard: streamlit run app/app.py · Code: github.com/TshinavheT/IDAproject (final_project)